

Theorem 1.1: formal statement and verified foundations

Falconer distance-gauge formalization project
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Status. This is an incomplete formalization notebook. The main theorem is an explicit proof obligation; it has not passed Comparator and is not registered on Palomar. The results below record the proved foundation.

The exact target

The uploaded manuscript defines

$$h_\theta(r) = r \exp\left(-(\log(1/r))^\theta\right) \quad (0 < r < 1), \quad h_\theta(0) = 0.$$

Its Theorem 1.1 asserts that, for a compact set $E \subseteq \mathbb{R}^2$,

$$\frac{2}{3} < \theta < 1, \quad \mathcal{H}^{h_\theta}(E) > 0 \implies \mathcal{L}^1(\Delta(E)) > 0.$$

$$\Delta(E) = \{\|x - y\| : x, y \in E\}.$$

The formalization uses the Euclidean norm and Mathlib's arbitrary-gauge Hausdorff measure constructor. For radii at least one, the gauge is extended by the identity; this does not affect its small-radius Hausdorff measure. The covering characterization has been proved:

$$\mathcal{H}^{h_\theta}(E) = \sup_{\rho > 0} \inf_{\substack{E \subseteq \bigcup_{n \geq 0} U_n \\ \text{diam}(U_n) \leq \rho}} \sum_{\substack{n \geq 0 \\ U_n \neq \emptyset}} h_\theta(\text{diam}(U_n)).$$

The independent `Challenge.lean` imports only Mathlib and repeats all custom definitions. The solution does not import Challenge. The theorem statement has no auxiliary Frostman, projection, or energy hypothesis.

The geometric foundation

Proposition 1. *For $0 < r < 1$, the gauge is positive and satisfies $h_\theta(r) \leq r$. Consequently $\mathcal{H}^{h_\theta}(E) \leq \mathcal{H}^1(E)$, and countable sets have zero gauge measure. Positive gauge measure supplies two distinct points of E .*

Proof. The exponential is strictly positive. The logarithm and its real power are nonnegative, so the exponential is at most one. Mathlib's gauge comparison theorem gives the measure inequality. Singletons are null for $\mathcal{H}^1$, hence also for the gauge measure; countable subadditivity gives the countable case. A set containing at most one point is countable. $\square$

The all-pairs distance map is continuous. It therefore maps the compact product $E \times E$ to a compact, Lebesgue-measurable distance set. Two distinct points give a positive distance; that fact alone does not give positive length.

The scalar summability foundation

Proposition 2. *For every $s, c > 0$, one has $\log n \leq cn^s$ for all sufficiently large natural numbers n . For every $\theta, c > 0$,*

$$\sum_{n=0}^{\infty} \exp(-cn^{\theta}) < \infty.$$

Proof. Mathlib proves $\log x = o(x^s)$ as $x \rightarrow \infty$ for $s > 0$. Restricting its eventual norm bound to natural numbers proves the first assertion. Applying that assertion with coefficient $c/2$ gives

$$\log n \leq \frac{c}{2}n^{\theta} \implies \exp(-cn^{\theta}) \leq \exp(-2 \log n) = n^{-2}.$$

The inverse-square series converges. The omitted finite initial segment has no effect on summability, so comparison proves the claim. $\square$

Proposition 3. *The branch-counting exponent and filter-error power obey*

$$3\theta - 2 > 0 \iff \theta > \frac{2}{3}, \quad \theta > \frac{2}{3} \implies -8(2\theta - 1) < -1.$$

Consequently, in the theorem's parameter range, every positive coefficient eventually absorbs the logarithm into $n^{3\theta-2}$, and

$$\sum_{n=0}^{\infty} n^{-8(2\theta-1)} < \infty.$$

Proof. The exponent inequalities are linear arithmetic. Apply the logarithmic domination result with exponent $3\theta - 2$, and apply Mathlib's real-power series criterion to the filter-error exponent. $\square$

These results justify scalar convergence once suitable shell energies and filter errors are supplied. They do not establish the geometric estimates that produce those bounds.

The exact rounded parameters

The gain has the proved exact form $\beta_N = c_{\theta}N^{\theta-1}$, with $c_{\theta} > 0$. The ceiling definitions are retained in the block parameter κ_N and tolerance ε_N . The verified lower bounds imply

$$\varepsilon_N N \geq \frac{(c_{\theta}/1000)^2}{10^6} N^{2\theta-1}, \quad \kappa_N^3 N \geq (c_{\theta}/1000)^3 N^{3\theta-2}.$$

Hence the actual rounded filter errors $(\varepsilon_N N)^{-8}$ are summable, and $\kappa_N^3 N$ eventually absorbs every fixed logarithmic cost when $\theta > 2/3$. The full Lemma 2.2 is now proved: all four budget groups hold above one common threshold. The coarse bound $T + 1 \leq 242N$ makes its derivative constants polynomial; logarithmic domination absorbs them into $2^{\varepsilon_N N/8}$. The remaining count budgets follow from the inverse tolerance bound and the growth of $\kappa_N^3 N$. All three terms in the asserted filter bound are now summable: $(\varepsilon_N N)^{-8}$, $N2^{-\varepsilon_N N}$, and $2^{-\kappa_N N/4}$. The geometric removed-mass estimate remains to be proved.

The verified gauge Frostman construction

Proposition 4. *For $0 < \theta \leq 1$ and a compact set E of positive gauge measure, there are a probability measure μ and a constant $C > 0$ such that*

$$\mu(E) = 1, \quad \text{supp } \mu \subseteq E, \quad \mu(\overline{B}(x, r)) \leq Ch_\theta(r) \quad (0 < r < 1).$$

Proof. Positive Hausdorff gauge measure gives positive gauge content. Finite dyadic trees supply nonnegative weights with uniformly positive total mass and saturated covering bounds. Choosing atoms on the compact set and normalizing gives probability measures carried by E . These measures form a weakly compact family, so a subsequence converges to a probability measure carried by E . Portmanteau transfers open-ball bounds to the limit. Comparing an arbitrary radius with consecutive dyadic scales, and applying the proved gauge doubling estimate, yields the closed-ball bound. The probability bound handles radii bounded away from zero. $\square$

Proposition 5. *The resulting measure has no atoms and has finite critical logarithmic energy for every $\gamma \geq 1$:*

$$J_\gamma(\mu) = \iint \frac{(1 + \log^+(1/\|x - y\|))^\gamma}{\|x - y\|} d\mu(x) d\mu(y) < \infty.$$

Proof. The ball bound tends to zero with the radius, proving atomlessness. The formal kernel is infinite on the diagonal. Decompose each pointwise potential into dyadic annuli; the gauge ball bound controls its n -th annular contribution by a constant times

$$(n + 1)^\gamma e^{-cn^\theta}.$$

Logarithmic domination absorbs the polynomial into half the stretched exponential. The resulting summable majorant uniformly bounds the potential. Integrating against the finite measure proves the energy bound. $\square$

The actual scalar Gaussian kernel is also controlled:

$$K_\gamma(r) = \sum_{n \geq 0} (n + 1)^\gamma 2^n e^{-4^n r^2} \leq C_\gamma \frac{(1 + \log(1/r))^\gamma}{r} \quad (0 < r < 1, \gamma \geq 0).$$

A split at the dyadic index nearest the inverse radius bounds the finite prefix and compares the shifted tail with $K_\gamma(1)$. This kernel now also controls the actual Fourier energy described on the next page.

The verified logarithmic Fourier energy

Proposition 6. *For every planar probability measure μ and $\gamma \geq 0$, a finite positive constant depending only on γ satisfies*

$$\int_{\mathbb{R}^2} |\widehat{\mu}(\xi)|^2 \frac{\log^\gamma(e + \|\xi\|)}{\|\xi\|} d\xi \leq C_\gamma(1 + J_\gamma(\mu)).$$

Proof. Mathlib's evaluated Gaussian Fourier transform and Fubini give the precise duality between Gaussian-weighted squared Fourier modulus and the spatial Gaussian kernel. Tonelli sums the non-negative dyadic Gaussian contributions. The proved scalar kernel bound controls their sum by the critical spatial energy. A dyadic scale nearest each high frequency supplies the desired logarithmic weight. At low frequencies, the probability bound controls the Fourier modulus; the extended inverse norm has a proved finite integral on the planar unit ball. Combining the regions proves Lemma 5.2 with a finite explicit constant and the manuscript's literal Fourier phase. $\square$

Separated probabilities and the prepared geometry

Two compact pieces of positive mass are extracted from the atomless Frostman probability. Their normalized restrictions retain gauge ball bounds and all finite logarithmic energies. The positive similarity used in Proposition 5.6 places the pieces in discs of radius $1/400$ inside the unit square, with centers exactly $1/4$ apart. Every cross distance lies in $[0.24, 0.26]$. These geometric and energy preparation statements are proved; the radial Orlicz density preparation remains to be established.

The actual product-distance pushforward has mass one on the all-pairs distance set. Its absolute continuity therefore gives positive length. Preparation multiplies this length by the positive similarity scale, so positivity transfers back to the original set.

The complete summable reconstruction criterion

Lemma 4.2 is now proved for a finite positive measure α and dominated approximants $\tau_N \leq \alpha$ for $N \geq N_0$. Its exact hypotheses are

$$\sum_{N \geq N_0} (\alpha(\mathbb{R}) - \tau_N(\mathbb{R})) < \infty, \quad \sum_{N \geq N_0} \int_{2^{N-1} \leq |r| \leq 2^{N+1}} |\widehat{\tau}_N(r)|^2 dr < \infty.$$

They imply $\alpha \ll \mathcal{L}^1$. The concrete convolution errors and Fourier bands form convergent L^1 and L^2 series. Telescoping the low-pass convolutions identifies their pairing with the original measure. The final absolute-continuity argument is given on the next page.

The verified reconstruction bridges

Dyadic frequency shells overlap at most three times. The quadratic L^2 bound proves convergence of square-summable shell pieces. Actual smooth low-pass kernels preserve mass; their differences have uniformly bounded L^1 norm and the precise dyadic Fourier support. Convolving removed measures with these differences produces summable L^1 errors whenever the removed masses are summable. Low-pass convolutions converge to the original measure on Schwartz tests.

Proposition 7. *Suppose a finite positive measure α satisfies*

$$\int \varphi d\alpha = \int \varphi f dx + \int (\mathcal{F}^{-1}\varphi)g d\xi \quad \text{for all Schwartz tests } \varphi, \quad f \in L^1, \quad g \in L^2.$$

Then α is absolutely continuous with respect to Lebesgue measure.

Proof. Let $d\sigma = |f| dx$. Cauchy–Schwarz and Plancherel give

$$\left| \int \varphi d\alpha \right| \leq \int |\varphi| d\sigma + \|g\|_2 \|\varphi\|_2.$$

Use smooth compact cutoffs equal to one on a compact subset of an open set U , zero off U , and bounded between zero and one. Their L^2 norm is at most $|U|^{1/2}$. Inner regularity yields

$$\alpha(U) \leq \sigma(U) + \|g\|_2 |U|^{1/2}.$$

For a Lebesgue-null set, outer regularity provides open covers whose Lebesgue and σ masses both tend to zero. The bound forces its α mass to be zero. $\square$

The actual L^2 shell pieces and the telescoped series pairing have been constructed, completing this part of the reconstruction proof.

Verified projection foundations

The orthogonal-projection development constructs jointly Borel nonnegative densities for almost every direction when the weighted Fourier energy is finite. Their fibers are the actual projection densities and satisfy the exact Plancherel identity. A smooth low-pass decomposition bounds squared density above an amplitude threshold by the corresponding Fourier tail. Summing logarithmic amplitude layers proves the literal one-dimensional bound

$$\int_{\mathbb{R}} f(t)^2 \log^\gamma(e + f(t)) dt \leq C_\gamma \int_{\mathbb{R}} |\widehat{\nu}(r)|^2 \log^\gamma(e + |r|) dr, \quad \gamma \geq 1,$$

for the actual density of a probability with square-integrable Fourier transform. No extra logarithmic exponent is introduced. Polar Fourier integration now gives the full averaged Lemma 5.3:

$$\int_{S^1} \int_{\mathbb{R}} F(w, t)^2 \log^\gamma(e + F(w, t)) dt dw \leq C_\gamma E_\gamma(\mu).$$

The jointly Borel density is the actual orthogonal projection density for almost every direction.

For radial projections, the inverse-distance-weighted smooth source has an exact ray-density identity with respect to circle arc length. The polar Jacobian cancels the inverse-distance factor. Extending rays to whole lines and applying the orthogonal estimate gives the smooth Orlicz transfer.

Uniform logarithmic bounds give an actual quantitative cutoff estimate:

$$\sigma_n(A) \leq H\lambda(A) + \frac{K}{\log^p(e + H)}, \quad p > 0.$$

Portmanteau transfers this estimate to a weak probability limit. Outer regularity extends it from open sets to arbitrary sets. Letting the cutoff grow proves absolute continuity of the limit. A uniform moment of order two greater than the desired exponent also gives finite lower Orlicz moments for the limit's actual Radon–Nikodym density. The target-specific radial endpoint and uniform spread preparation are now assembled below.

Verified regular decomposition

Lemma 5.8 is proved for the original measure weights. The half-open unit square has the exact terminal-cell count. The construction discards light terminal cells, assigns bottom-up types using original masses, and discards whole light type fibers. The rounded schedule is exact:

$$\Delta = \lfloor \varepsilon N / 4 \rfloor, \quad w = \lfloor \varepsilon N / 8 \rfloor, \quad k = \lceil N / \Delta \rceil.$$

Actual parent-child counts bound each class drop, yielding ten possible choices per sampled level. The type count and (P2) give

$$\#\{\text{types}\} \leq (48/\varepsilon + 1)10^{8/\varepsilon+1} \leq 2^{\kappa N/4}.$$

The retained Borel carriers are disjoint unions of original positive-mass terminal cells, with the exact estimates

$$\mu(G_t) \geq 2^{-\kappa N/2}, \quad \mu\left([0,1)^2 \setminus \bigcup_t G_t\right) \leq 2 \cdot 2^{-\kappa N/4}.$$

Restriction preserves each retained terminal-cell weight. Every coarser restricted cell mass equals its original-weight fiber sum. The sampled-level comparisons interpolate to every depth through N . Normalization multiplies all masses by one factor, so the actual normalized probability satisfies

$$\rho_t(Q) \leq 2^{\varepsilon N} \rho_t(P), \quad 0 \leq n \leq N, \quad P, Q \text{ depth-}n \text{ cells}, \quad \rho_t(P) > 0.$$

This is the source's regularity condition under its finite-depth convention.

Verified maximal-cell excess bounds

The maximal cell mass is an actual attained maximum over the finite cells of the unit square. For every probability carried there, successive values obey

$$M(0) = 1, \quad M(n+1) \leq M(n) \leq 4M(n+1), \quad M(n) > 0.$$

The upper successor bound comes from the four actual children of a parent cell; the lower one follows because every child lies in its parent. The excess function is defined from these actual masses by

$$M(n) = 2^{-n} 2^{-NA(n)}.$$

The successive ratios lie in $[1, 4]$, so their base-two logarithms lie in $[0, 2]$. This proves the exact Lipschitz constant for A .

For a retained carrier G , its original mass is at least $2^{-\kappa N/2}$. Every dyadic cell fits in a ball of radius exactly its side length about its midpoint, so its original mass is bounded by $Ch_\theta(2^{-n})$ without an extra radius factor. Normalizing gives

$$M(n) \leq C 2^{\kappa N/2} 2^{-n} 2^{-Ng(n)}.$$

The constant budget $\log_2 C \leq \kappa N/2$ yields the lower excess estimate. All four conclusions of Lemma 5.10 are proved for the actual restriction:

$$A(0) = 0, \quad |A(n) - A(m)| \leq \frac{|n - m|}{N}, \quad g(n) - \kappa \leq A(n) \leq \frac{n}{N}.$$

The upper bound follows from the Lipschitz bound and $A(0) = 0$.

Verified radial endpoint and uniform spread

The gauge supplies every logarithmic energy order. Smooth compact probability approximations contract their Fourier energies and stay in a common carrier separated from the pins. The averaged orthogonal estimate and the exact smooth polar identity bound the actual joint radial densities at a stronger Orlicz order. Their joint measures converge weakly; the proved Orlicz limit argument identifies the actual Radon–Nikodym density of the limit. Thus the prepared radial projections are absolutely continuous and have finite moments at every desired order for almost every pin. This target-specific argument uses stronger energies than the general single-order Theorem 5.4.

Write the eighth-order radial moment as

$$Q(x) = \int_{S^1} \Phi_8(F_x(w)) dw, \quad \Phi_8(t) = t \log^8(e + t).$$

Joint measurability makes the good-pin sets with bounded moment Borel. Since these sets exhaust a full-measure set, one has mass greater than one half. Inner regularity gives a compact subset with original mass greater than one quarter. Restrict and normalize both sources to these compact sets. Each restricted radial measure is at most four times its original radial measure; the actual Radon–Nikodym densities satisfy the same domination almost everywhere. The elementary scaling estimate is proved:

$$\Phi_8(4t) \leq 4(1 + \log 4)^8 \Phi_8(t), \quad t \geq 0.$$

A common finite bound therefore holds at **every** retained pin in both directions. The new probabilities retain the small separated discs, actual compact carriers, and all logarithmic energies. Their common gauge constant increases by at most four. This proves the uniform spread requirement of Proposition 5.6(c), with the actual radial densities.

The logarithmic small-set estimate is also proved for actual densities:

$$\int_Z F \leq H|Z| + K(\log H)^{-8}, \quad H > 1.$$

For the manuscript’s bad-direction length budget, choosing its inverse square root gives the exact bound by a dyadic exponential plus an inverse eighth power. Actual test geometry and filtered-measure loss are still required before this becomes Lemma 6.13.

Verified entry depth and finite test schedule

Actual regular-part cell masses now give the source's occupied-descendant count, rather than a postulated counting bound. The entry depth is the last integer depth at which the actual excess is at most the gain:

$$c = \max\{n \leq N : A(n) \leq \beta\}.$$

The exact parameter facts and gauge lower barrier prove all the entry bounds of Lemma 9.2, including the post-entry budget:

$$\lfloor \beta N \rfloor \leq c < N/4, \quad 100q < N - c, \quad A(n) > \beta - 1/N \quad (c \leq n \leq N),$$

$$V([c, N]) \geq 2\beta - 2/N, \quad k_c[0, c] \leq \beta + \kappa.$$

The finite interval minima define the actual trimmed budget. Its inclusion monotonicity and split gains are proved from actual minimum plateaux. The constructed split uses marked block minimizers and gives the length margins that pay the positive costs of the three moves.

Contributing tube and projection tests are concrete finite records with anchors, endpoints and lengths. Their scheduled lists run over the actual long subintervals, are monotone under interval inclusion, and retain each contributing origin. Anchor depths lie in the constructed marks and every length is at most half its origin's length. The mark count and the exact parameter budget give at most N^2 tests per list, proving Lemma 8.3(b,c). This is finite schedule arithmetic; the analytic move estimates and chain induction remain to be proved.

Verified uniform nonstationary estimates and smooth bumps

For a smooth amplitude with derivative bounds

$$\|a^{(j)}\|_\infty \leq AM^j \quad (j \leq K), \quad M \geq 1,$$

assume the actual phase derivative has norm at least a positive number on an open neighborhood of the amplitude's closed support, and its higher derivatives through order $k + 1$ have norm at most a common constant. A globally smooth regularized reciprocal agrees with the true inverse there. Exact Leibniz coefficients and factorial bounds give the source's polynomial base:

$$\left| \int e^{i\phi} a \right| \leq |\text{supp } a| A \left(\frac{2(k+1)^2 \max\{M, \Lambda/\lambda\}}{\lambda} \right)^k.$$

The smooth compact real-line form and the smooth periodic form are both proved. For the latter, endpoint terms cancel over an actual period, and its length replaces the support length. For a linear phase, the transported amplitude is exactly the k -th derivative times the k -th power of the constant inverse slope; the sharper base M divided by the slope lower bound allows M below one. No unspecified order-dependent derivative constants are used.

For every finite order K there is a genuine smooth, nonnegative, even probability bump with compact support in $[-1, 1]$ and

$$\int_{\mathbb{R}} |\omega_K^{(k)}(x)| dx \leq (\pi^2/6)^k (k!)^2, \quad 0 \leq k \leq K.$$

Its construction convolves normalized boxes of half-widths $6/(\pi^2 j^2)$, with a smooth probability tail inside the remaining support radius. The fundamental theorem of calculus differentiates each box average into its boundary difference. The L^1 cost is exactly the inverse box width, giving the factorial product. The Basel identity makes the remaining radius positive. This finite-order construction suffices for the later finite-order masks; an infinite-product identity is not claimed.

Remaining analytic work and verification

The proof still needs actual directional tests and masks, filtered-measure mass estimates, stationary phase, four uniform energy estimates, and the multiscale budget induction. Fixed-parameter estimates need proved uniform dependence before use at shrinking parameters.

The main theorem currently has one explicit proof hole. Comparator permits the three standard axioms and must reject the target's proof hole. A successful ordinary build of this skeleton is not a successful theorem verification.

The public project includes a separate sandboxed Comparator job and a manual full Palomar preflight. Registration requires a completed proof, verification of the exact public commit, truthful metadata, and the registry review.

[Public project: CoolRmal/falconer-theta-gauge](#)